

PROJECT REPORT: STUDENT MANAGEMENT SYSTEM

Department of Artificial Intelligence Riphah International University

Course: Object-Oriented Programming (CS-213)

Semester: 2

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1. SUBMISSION INFORMATION

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2. PROJECT ABSTRACT

The **Student Management System (SMS)** is a Java-based desktop application designed to streamline educational administration. By implementing a centralized data structure, the system manages student enrollment, faculty assignments, grading logic, and financial records. The project is built entirely using **Object-Oriented Programming (OOP)** to ensure the code is secure, reusable, and easy to maintain.

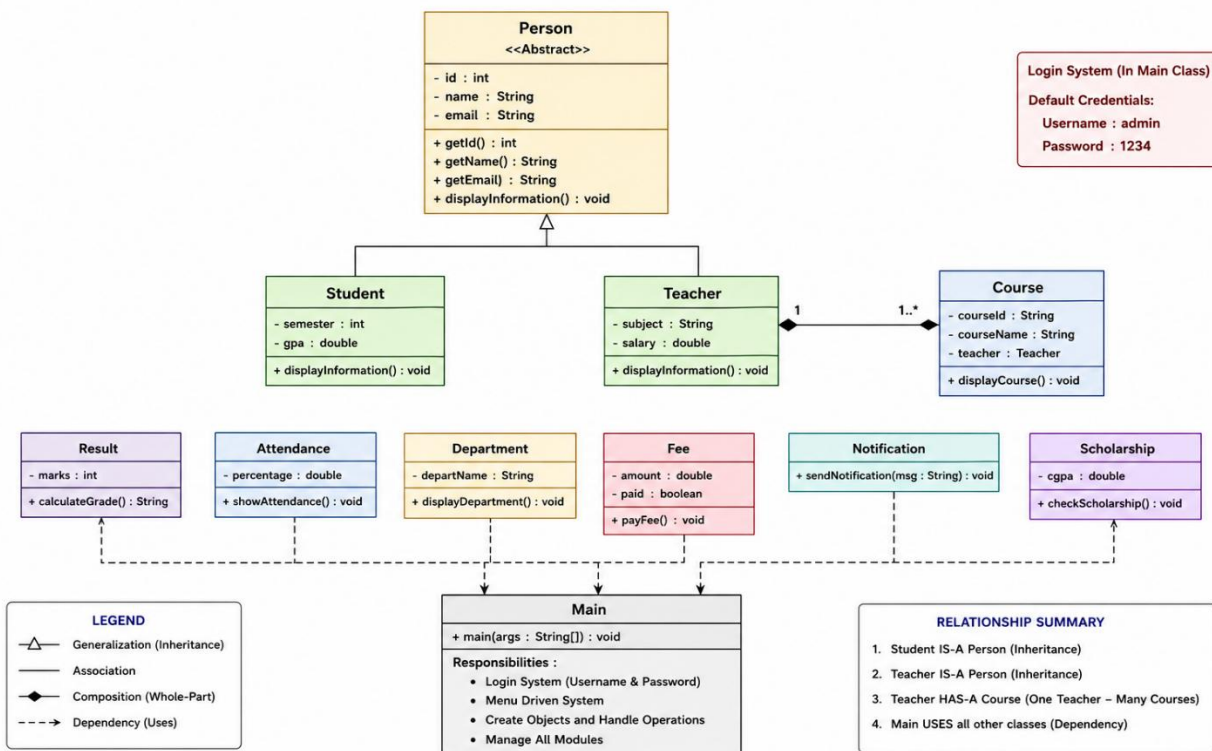
3. SYSTEM DESIGN & ARCHITECTURE

The system architecture follows a modular hierarchy. The design ensures that human entities, academic results, and financial transactions are handled by separate, interacting classes.

3.1 UML Class Diagram

The logic is centered around a hierarchical structure. The Person class acts as the blueprint, while the Student and Teacher classes inherit its properties.

UML Class Diagram – Student Management System (Final)



4. IMPLEMENTATION OF OOP PRINCIPLES

- **Inheritance:** The Student and Teacher classes inherit common attributes like Name, ID, and Email from the base Person class, which significantly reduces code duplication.
- **Abstraction:** We defined the Person class as abstract to prevent the creation of generic "Person" objects. It forces the implementation of the displayInformation() method in all child classes.
- **Encapsulation:** Sensitive variables like GPA, Salary, and Fee Status are kept private. They can only be accessed or modified through public "Getter" and "Setter" methods.
- **Polymorphism:** Through method overriding, the system uses a single method call (displayInformation()) to show different data formats for students and faculty.

5. CORE FUNCTIONALITIES

1. **Administrative Security:** A login wall requiring valid admin credentials (admin/1234).
 2. **Financial Module:** Automated billing that validates payment status to prevent duplicate transactions.
 3. **Academic Grading Engine:** * **Automated Grading:** Assigns grades (A+, A, B, etc.) based on marks entered.
 - **Scholarship Logic:** Calculates tuition waivers (up to 100%) automatically based on the student's CGPA.
 4. **Notification System:** A dedicated service for broadcasting administrative messages to the console.
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6. CONCLUSION

This Student Management System successfully translates theoretical OOP concepts into a functional software tool. The current console-based version provides a solid foundation for future expansion, such as adding a Graphical User Interface (GUI) or connecting to a permanent SQL database.

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